

Preface

This author is no genius, but wrote the "Machine Learning" series before this "Math Made Easy" series; the two series both belong to the same "Iris Book" learning framework.

"Machine Learning" series — all 7 volumes — set out to build an engaging, substantive, and rewarding "math + programming + visualization + machine learning" learning ecosystem for the coming "quasi-artificial-intelligence era." The whole series has since been published and has won a great deal of recognition from readers — which, if anything, has left this author all the more anxious and walking on thin ice.

Why Write the "Math Made Easy" Series?

Given that the "Machine Learning" series already exists, why start over from scratch?

Simply put, five years ago, starting from when the first manuscript of the "Machine Learning" series was open-sourced on GitHub, even though this author had already tried to lower the vantage point as much as possible, many "zero-background" readers kept reporting that the math portions of the "Machine Learning" series were "too hard" (though of course, some experts said it was too simple). Beyond that, many readers reported that the "Machine Learning" series, at 7 volumes, is such a sprawling system — spanning math all the way to machine learning — that the most common obstacle is the foundational math concepts.

"Math Made Easy" is a new work focused on the learning experience of readers with zero background. Each volume of "Math Made Easy" tries to break foundational concepts down piece by piece and explain them with total clarity, doing everything possible to make the material more learnable.

"Math Made Easy" is not just a math-foundations study companion for the "Machine Learning" series — it carries the essential mission of helping learners at every stage build a solid mathematical foundation:

- ▶ It's suited to helping high-school students shore up the math foundations they need for informatics olympiads.
- ▶ "Math + programming + visualization" helps undergraduates appreciate the beauty of mathematics while also putting what they learn to practical use.
- ▶ For those returning to retrain after graduation, it strings together a network of knowledge, deepens understanding, and builds sturdier mathematical thinking.
- ▶ It offers a clear approach for graduate-entrance-exam math preparation, making it easier to master mathematical tools and lightening the study burden.
- ▶ It helps programming learners and programmers understand the mathematical principles behind programming.
- ▶ It's a great helper for learning math from zero and getting started with programming, letting even humanities students sweep up the core knowledge in one go, easing anxiety, and making it worry-free to cross into a new field.

AIGC as Your Study Companion, Start to Finish

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AI's rapid development has already turned large language models (DeepSeek, ChatGPT, and others) into learning's good teachers and helpful friends. As tools, large language models are penetrating every field at an unprecedented pace — from text generation and code writing to scientific research and financial analysis, their reach is nearly everywhere. In education, large language models can organize knowledge, build personalized learning paths, and also provide real-time answers, visualize complex concepts, and enhance the interactivity of learning.

AI-companioned study is already an educational trend, and it carries enormous practical significance.

"Math Made Easy" also tries to make large language models part of its teaching ecosystem, and attempts to guide readers toward carrying this AI-companioned way of learning deep into their future study habits.

"Math Made Easy" Is Different From the "Machine Learning" Series

"Machine Learning" fuses together the key elements of "programming + visualization + math + machine learning," and its learning endpoint is machine learning itself; "Math Made Easy," by contrast, emphasizes a thorough understanding of mathematical concepts and applying them as tools — its starting point and focus are both mathematics.

"Machine Learning" is a sweeping, 7-volume machine learning system; "Math Made Easy," instead, focuses on the math domain (linear algebra, calculus, and probability and statistics) as its unit, staying "light and focused."

In terms of mathematical difficulty, "Math Made Easy" is better suited to zero-background readers — simpler and more basic; but on the network of foundational math knowledge, "Math Made Easy" is actually more comprehensive.

"Machine Learning" particularly emphasizes the "programming ability" within "visualized programming"; the "programming + visualization" in "Math Made Easy" is merely a means for breaking down and implementing mathematical concepts, not an end in itself.

"Math Made Easy," compared with the "Machine Learning" series, rewrites its content entirely even for the same knowledge points, because the problems it needs to solve are different; the vast majority of the images have been recreated from scratch, and the visualization schemes are more streamlined.

Readers who have already studied the "Machine Learning" series and found it entirely smooth sailing don't need to read "Math Made Easy"; but if it felt like an endless range of mountains, you can use "Math Made Easy" to clear away the obstacles and then go back to the "Machine Learning" series — you're sure to find it suddenly makes sense.

Readers who first study "Math Made Easy" to build a solid foundation can, if needed, go on to study the "Grand Mathematics" series and move toward machine learning algorithms and applications — this is where the productivity generated by math + programming truly lies.

AI Era: Math Isn't Useless — It's Becoming More Important

AI tools have become so widespread that they've made a lot of work simpler and more efficient. We need to think about one question: if AI can do so many things, where will our future professional competitiveness come from?

A real-world scenario: a company laid off 60% of its programmers, keeping mainly architects and algorithm engineers. In the past, they led teams to collaboratively finish development work; now, they work side by side with AI, getting tasks done efficiently.

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Mathematics is the foundation for understanding the world, an important tool for solving problems, and the underlying logic that drives red-hot large language models like DeepSeek and ChatGPT. Faced with increasingly intelligent large language models, we not only need to learn how to use them, but also need enough mathematical literacy to understand their principles, recognize their limitations, and avoid being fooled or misled.

With DeepSeek and ChatGPT around, nobody needs to memorize theorems and formulas by rote anymore!

Understanding, not rote memorization. Only by mastering a solid mathematical foundation can we hold an "equal" conversation with these intelligent tools, fully unlock their potential, and boost our work and study efficiency. In this age of intelligence, mathematics is no longer just subject knowledge — it's a core competitiveness everyone should possess.

Leave "math for math's sake" to the "geniuses"; letting "ordinary people" understand and use the common tools in the math toolbox equally and efficiently — that's what "Math Made Easy" is after.

For "ordinary people," whether in STEM or the humanities, mastering linear algebra (plus some matrix theory), calculus (single- and multivariable calculus, plus optimization methods), and probability and statistics (univariate and multivariate) is enough to handle most mathematical application scenarios.

In short, one's "grasp" of foundational math knowledge is one of the decisive factors determining how high a career ceiling can go.

Programming Isn't the Goal — It's a Tool, a Means

Today, with artificial intelligence advancing rapidly, programming has long since stopped being just a programmer's skill — it has become a "second language" everyone should master. Just like mastering a foreign language, programming has become a "required course"! Just as English was once the passport of the globalization era, programming is becoming an essential tool of the data era.

In "Math Made Easy," the goal of learning to program isn't to turn everyone into a "code monkey," but to solve real-world problems more efficiently — to let theory from math, science, engineering, finance, and other fields truly land, turning it into an executable plan. Even when using DeepSeek, ChatGPT, and other tools to generate code, you still need to state your requirements clearly, understand what the code does, know how to modify and optimize it, and master how to deploy it.

AI tools have indeed raised programming efficiency, but they've also raised the bar for programming ability, requiring users to have logical thinking, problem-analysis skills, and the ability to debug code.

"Math Made Easy" also tries to help everyone get started with programming, sharpen their programming skills, and solve concrete problems.

The Solution: Programming + Math

This author believes that learning math while using Python to program and verify calculations at the same time is an efficient plan for "ordinary people" to learn both math and programming.

Programming lets mathematical tools land in the real world, shortening the distance from theory to practice; mathematics gives programming its logic and depth, expanding the boundaries of technology.

Mathematics provides rigorous logical thinking and powerful computing ability, while programming languages such as Python let these mathematical tools be quickly implemented.

Especially during self-study, "**learn math while using Python to compute**" is a highly efficient method — it not only helps deepen everyone's understanding of mathematical concepts, but also builds practical ability, so that knowledge no longer stays on the page but becomes a tool that can be verified and put to use.

The Optimal Solution: Programming + Math + Visualization

Visualization is the Iris Book "Machine Learning" series' sharpest weapon! And of course, visualization is "Math Made Easy"'s signature trick too!

Mathematics isn't just a series of symbols and formulas — it's an abstract expression of the world's underlying patterns.

The real, physical world is a geometric world. Human beings are visual learners, and complex concepts should be presented through visualization to make the learning process more intuitive and easier to understand. This author believes there's no better way to learn than by visualizing abstract mathematical concepts, so that every formula can form an intuitive geometric picture in the mind.

"Math Made Easy" equips every topic with a large number of high-quality flat visualization schemes, along with visualization videos. This "math + programming + visualization" way of learning doesn't just dramatically lower the difficulty of understanding — it leaves a deep impression of the knowledge in the brain, truly achieving "never forget once seen."

Compared with rote memorization, mathematical visualization lets us grasp core concepts intuitively, making it easier to use mathematical tools to solve real problems. Mastering this way of learning means you're not just learning math — you're training your mind, turning every formula into a powerful weapon for solving problems.

 Note that not every reader needs to master complex visualized programming; so while "Math Made Easy" provides a large number of visualization schemes, its programming portion emphasizes "computation," not drawing. Readers who need "visualized programming" can refer to the "Machine Learning" series.

Thanks

Special thanks to my students and reader friends for their trust! The open-sourcing and publication of the "Machine Learning" series received tremendous support from a great many students. Many students left messages or wrote emails telling this author that they had gone through the "Machine Learning" series from cover to cover many times over, running the companion code more than once, and gained a great deal from it. Many students also offered critical feedback. In this author's eyes, these interactions are more precious than any "award." They not only prove the value of this set of books, but are also the biggest driving force behind this author's continued updates to the content, refinements to the visualization schemes, and polishing of every detail.

Special thanks to Mr. Luan Dacheng of Tsinghua University Press for accompanying this journey all the way through. Without exaggeration, without Mr. Luan Dacheng there would be no Iris Book "Machine Learning" series, and there would be no "Math Made Easy" either! Over these past few years, this author has chatted with Mr. Luan Dacheng "for a few minutes" almost every single day, on topics ranging from creative direction, structure, and content to layout, proofreading, printing, book merchandise, and the odds and ends of daily life and work ...

"Math Made Easy" has gained a new team member — Cui Ying (CuiCui)! CuiCui holds dual master's degrees from the Harbin Institute of Technology and Canada's McMaster University, and she is a longtime friend of mine who, like me, loves mathematical visualization. From programming,

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debugging, and finalizing materials to editing and post-production, CuiCui has taken charge of the entire production process for "Math Made Easy"'s companion videos.

Acknowledgement

To my parents.

How to Use This Book

"Math Made Easy": What Are Its Distinguishing Features?

If readers can't understand it, that's the author's problem. Reach the author directly through any media channel: tell me where it doesn't make sense, and I'll fix it!

No assumed threshold, no filtering. It pays special attention to the learning experience of zero-background readers, assuming the reader knows nothing at all.

Raising interest, satisfying curiosity, and generating intrinsic motivation are the goals.

Learning for the sake of using it, with application as the priority, using mathematical tools to solve concrete problems.

A scaffolded, heuristic method of learning. Knitting knowledge into a network, reinforcing understanding of the same concept from different angles at different points.

Not greedy for more, not chasing completeness — explaining knowledge thoroughly is what makes it complete.

Innovation, creation, and guided understanding, rather than rote memorization, grinding through problem sets, or racking up problem-solving tricks. Leave rote memorization to the language models.

Pleasing to the eye — math has to be beautiful. Practicing "math + art," using the image quality of Beauty of Data Visualization to dig out the geometric intuition behind "simple" math concepts.

"Math Made Easy" provides richer, more spectacular visualization schemes to set your imagination free. Every formula and every theorem is explained through images; the images are richly detailed, but every detail serves the knowledge point, ensuring you never forget it once you've seen it.

Dressing math up so that everyone loves it at first sight, keeping formulas to a minimum and explanations conversational.

"Math Made Easy" provides exercises in every section, offering an extensible learning framework. It offers a more efficient, harder-to-forget learning path. It combines the learning experience of large language models (DeepSeek, ChatGPT, and others) into the mix. Guided independent learning: the exercises use DeepSeek\ChatGPT to help extend your knowledge.

Free and open source! Continuing the fine tradition of the "Machine Learning" series. The PDF manuscript, Python code, and videos — all free and open source.

 A special note: although "Math Made Easy" covers, and is not limited to, most of the knowledge points in undergraduate math, it does not try to replace the traditional model of math instruction — it's meant only as a supplement, offering an alternative learning path, and there's no plan in the near term to produce teaching courseware. "Math Made Easy" will also recommend more advanced classic textbooks, paying tribute to those books.

"Math Made Easy": Who Is It For?

Written for those who find abstract mathematical symbols dull, but who are sensitive to geometric shapes.

Readers who find math "fun" and math "beautiful" — this is for you.

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Those who want to have a breakthrough, raise their curiosity, boost their thirst for knowledge, and improve their spatial imagination.

Readers who aren't satisfied with just "problem-set tactics and solving tricks".

Those picking math back up again. If you gave up on math, or were told to give up on math, why not give "Math Made Easy" a try!

Readers with a solid math background can focus on "Math Made Easy"'s aesthetic experience.

Readers who want to bring "math + programming + visualization" together, all three at once.

Readers who want to know what math is good for and how to use math.

Readers learning to program, but without a clear direction.

Companion Resources

"Math Made Easy" provides the following companion resources:

- ◀ A printed book;
- ◀ PDF manuscript, for convenient study on mobile devices; please note that the printed book has been revised through the publisher's five-round review-and-proofread process, so its content details may differ from the PDF file.
- ◀ Python code files (Jupyter Notebooks);
- ◀ Fun, substantive short videos that build spatial imagination and geometric intuition.

The download address for this series' PDF manuscripts and Python code files is:

- ◀ <https://github.com/Visualize-ML>

Python code files are revised from time to time, so please keep an eye out for updates. The book's companion PDF manuscript is also uploaded to this GitHub account. It's therefore recommended that everyone register a GitHub account and star the manuscript folder (star) or fork it (fork).

Readers who want to focus specifically on learning Python programming are advised to study Python for Beginners with the help of tools such as DeepSeek/ChatGPT. The download address for the manuscript and companion Python code is

- ◀ https://github.com/Visualize-ML/Book1_Python-For-Beginners

Readers who want to sharpen their visualized programming are advised to study Beauty of Data Visualization with the help of tools such as DeepSeek/ChatGPT. The download address for the manuscript and companion Python code is

- ◀ https://github.com/Visualize-ML/Book2_Beauty-of-Data-Visualization

Readers who want to learn machine learning algorithms are advised to study First Course in Data Science and Visualizations for Machine Learning with the help of tools such as DeepSeek/ChatGPT. The download address for the manuscripts and companion Python code is

- ◀ https://github.com/Visualize-ML/Book6_First-Course-in-Data-Science
- ◀ https://github.com/Visualize-ML/Book7_Visualizations-for-Machine-Learning

Readers who want to firm up the mathematical foundations of machine learning are advised to study Elements of Mathematics, Power of Matrix, and Essentials of Probability and Statistics with the

help of tools such as DeepSeek/ChatGPT. The download address for the manuscripts and companion Python code is

- ◀ https://github.com/Visualize-ML/Book3_Elements-of-Mathematics
- ◀ https://github.com/Visualize-ML/Book4_Power-of-Matrix
- ◀ https://github.com/Visualize-ML/Book5_Essentials-of-Probability-and-Statistics

Practice Platform

The IDE (integrated development environment) this book's author used to write the code is Spyder, with the goal of giving everyone clean, simple Python code files.

That said, it's recommended that everyone use JupyterLab or Jupyter notebook as the companion study tool for this series.

Simply put, Jupyter brings together "browser + programming + documentation + plotting + multimedia + publishing" — all these capabilities in one, and it's extremely well suited to inquiry-based learning.

Running Jupyter needs no IDE — just a browser. Jupyter makes it easy to execute code block by block. Jupyter supports inline result printing, printing the resulting images directly below the code block. Jupyter also supports many other languages, such as R and Julia.

Using its markdown document-editing feature, you can write notes while you program, with no need to create a separate document. In Jupyter, inserting images and video links is easy. You can also insert Latex formulas. For long documents, you can use the sidebar table of contents to find specific content.

Jupyter's publishing feature is very friendly, making it easy to print to HTML, PDF, and other formats.

You can download and install Anaconda; JupyterLab, Spyder, PyCharm, and other common tools are all integrated within Anaconda. The download address for Anaconda is:

- ◀ <https://www.anaconda.com/>

For installing Anaconda and using JupyterLab, please refer to the first three chapters of Python for Beginners; the download address is

- ◀ https://github.com/Visualize-ML/Book1_Python-For-Beginners

Comments and Suggestions

Comments and suggestions about this series are welcome; the series' dedicated email address is:

- ◀ jiang.visualize.ml@gmail.com

The "Iris Book" has always kept up one tradition: for readers who offer key comments and valuable suggestions, we prepare small pieces of merchandise (an Iris-branded umbrella, a summit backpack, a shoulder bag ...) as a small token of our appreciation.

Contents

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Introduction

Seeing the Geometric Beauty Behind the Formulas

0.1 A Chat About Linear Algebra and Linear Algebra Made Easy and Visual

Simply put, linear algebra is the branch of mathematics that studies vectors, matrices, and their linear-transformation relationships.

Linear algebra is the mathematical language of modern science and engineering. It's not only the bridge from single-variable calculus to multivariable calculus, but also the core tool for moving from univariate probability and statistics to multivariate statistical analysis. In cutting-edge fields such as machine learning, computer vision, natural language processing, robotic control, image processing, quantitative finance, and data analysis, linear algebra is almost everywhere.

Linear algebra is a foundational course commonly offered at the undergraduate level. This author believes the core mission of undergraduate education is to train qualified professionals, not to "filter" or "weed out" students; the high-school-style teaching model oriented around "rote memorization," "drilling problem sets," and "scores" is therefore no longer suited to higher education.

Linear algebra instruction should not, therefore, become an artificially erected barrier — it should become an important starting point for helping students build modern scientific thinking. Every college graduate should be able to wield linear algebra as a comfortable, well-worn tool in their toolbox, applying it freely in future study, research, and work.

This also raises the bar for how teachers teach, how courses are designed, and how textbooks are organized. Linear algebra instruction shouldn't stop at piling up symbols and deriving formulas — it should combine graphical intuition, real applications, and programming/computational tools to help students understand the geometry and structure behind vector spaces, linear transformations, and matrix operations. Modern teaching can make full use of visualization, hands-on programming, and modeling case studies, letting students "see the beauty of mathematics" and "put math to living use," thereby sparking their interest in learning and their ability to apply it.

This author believes today's linear algebra instruction genuinely has some problems. Specifically:

- ▶ It neglects geometric intuition. Two-dimensional/three-dimensional images are rarely used to show abstract concepts, leaving students unable to "understand it or even imagine it." For example, many textbooks introduce the determinant right at the start, with almost no explanation of its geometric meaning.
- ▶ It emphasizes calculation over understanding. Practice problems lean toward routines, tricks, and oddball difficult problems, neglecting understanding of geometric essence and the cultivation of

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modeling ability. Many textbooks follow a "definition-theorem-example" pattern that's too mechanical, focused only on "answering the question" rather than on "truly understanding" or "being able to use it."

- ▶ It doesn't emphasize connecting theory with practice. There's a lack of connection to real problems, the real world, or other disciplines. Most textbooks barely mention real-world application cases of linear algebra, such as data analysis, image processing, machine learning, graph theory, and networks.
- ▶ Instruction is disconnected from modern tools. Teaching needs to be combined with Python programming, visualization, and animation. Most textbooks are pages full of formulas, without actively combining graphics, Python programming, and visualization tools to deepen understanding and build application scenarios.

Linear Algebra Made Easy and Visual tries to solve these problems.

- ▶ Systematically introducing geometric intuition. It uses large numbers of graphics and animations to show the geometric intuition behind concepts such as vectors, matrices, matrix multiplication, linear transformations, determinants, inverse matrices, linear equations, eigenvalue decomposition, and singular value decomposition — even trying to let readers glimpse "the beauty of mathematics"!
- ▶ Shifting the focus from "solving problems" to understanding and application. By introducing real applications such as principal component analysis, image compression, and least-squares fitting, it lets readers truly see the practical use of mathematical tools, truly achieving "what's learned gets used."
- ▶ Reasonably designing the knowledge structure. It starts with vectors and matrices, rounds out core tools such as matrix multiplication, determinants, inverse matrices, and vector spaces, and only then moves deeper into advanced topics such as solving systems of linear equations, linear transformations, eigenvalue decomposition, and singular value decomposition — with application cases for linear algebra tools woven in throughout.
- ▶ Learning math, programming, and visualization side by side. It folds Python programming, Jupyter Notebook, interactive graphical tools, and large language models into instruction, letting students truly understand math while "writing code," and build modeling instincts while "visualizing hands-on."

0.2 The Structure of This Book

Linear Algebra Made Easy and Visual is split into two volumes. Volume 1's topics include relatively foundational material such as vectors, matrices, matrix multiplication, determinants, inverse matrices, vector spaces, and systems of linear equations; Volume 2, meanwhile, covers more advanced linear algebra, matrix theory, and data-analysis/machine-learning topics, such as linear transformations, eigenvalue decomposition, singular value decomposition, principal component analysis, least-squares fitting, the Rayleigh quotient, matrix norms, and discriminant analysis.

This book is Volume 1 of Linear Algebra Made Easy and Visual, comprising 7 chapters in total; below is a brief introduction to the core content of each chapter.

Vector

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Vectors are arguably the most geometrically intuitive concept in linear algebra, which is why we've placed vector-related topics in this book's first chapter. This chapter starts by introducing vectors from an algebraic angle, bringing in RGB color as an example that runs through the whole chapter. We then place vectors within a coordinate system and start building geometric intuition for them. Next, we introduce vectors' various common operations, such as addition and subtraction, scalar multiplication, normalization, the inner product, projection, and norms. Every one of these topics is introduced hand in hand with geometric pictures.

Matrix

Row vectors and column vectors are both special cases of matrices. Naturally, Chapter 2 turns to introducing matrices. It starts with matrices' common properties, shapes, and basic operations. Please pay particular attention to matrix shape, since so many linear algebra operations require a specific matrix shape as a precondition. Matrix transpose is simple, but it's also one of the topics people most often get wrong, so this chapter devotes an entire section to explaining it. The chapter then spends three whole sections on a first look at matrix multiplication: we first focus on the rules of matrix multiplication, then use four sets of examples to explain the geometric transformations embedded within it; once you have this geometric-transformation lens, learning the properties of matrix multiplication becomes especially easy.

Matrix Multiplication

Matrix multiplication plays an absolutely central role in the linear algebra framework. That's why we've devoted an entire chapter to it. This chapter focuses on the different perspectives on matrix multiplication — the inner product, the outer product, linear combinations of column vectors, and linear combinations of row vectors. The inner-product view is the most straightforward lens for understanding the rules of matrix multiplication. The outer-product view will go on to play an important role in later topics such as spectral decomposition, singular value decomposition, and principal component analysis. The column-vector and row-vector linear-combination views are, in effect, laying the groundwork for topics such as vector spaces, systems of linear equations, and linear transformations. The chapter closes by giving the whole topic a further "lift" from the perspective of block matrix multiplication.

Determinant

Unlike many textbooks that lead with the determinant right away, this volume places it in a relatively later position. This book treats the determinant as a tool used in applications such as computing inverse matrices and solving systems of linear equations. When explaining the determinant, though, this book focuses on its geometric intuition. The chapter's first section introduces the relationship between the determinant of a 2×2 square matrix and the area of a parallelogram; in this section we'll see the parallelograms corresponding to different types of square matrices. The second section further "lifts the dimension," introducing the relationship between the determinant of a 3×3 square matrix and the volume of a parallelepiped. With this geometric groundwork in place, understanding the properties of the determinant becomes especially easy. The chapter closes by introducing Laplace expansion for computing determinants.

Inverse Matrix

A geometric lens makes understanding the inverse matrix especially direct! Simply put, the inverse matrix is the inverse transformation that undoes a geometric operation. If the geometric operation corresponding to a matrix multiplication has an inverse operation, the matrix is invertible; otherwise, it isn't. This chapter is organized around exactly that idea. We first introduce the inverse matrix using geometric transformations such as scaling, rotation, and shear; then, again using a geometric lens, we explore the properties of the inverse matrix. The chapter closes by introducing how to compute the inverse matrix using the adjugate-matrix method, while also revisiting Laplace expansion for computing determinants.

Vector Space

With the groundwork laid by the earlier topics of vectors, matrices, matrix multiplication, determinants, and inverse matrices, understanding the core concepts of vector space becomes especially easy. This chapter again uses the RGB color-space example. Note that the RGB color space clearly isn't a vector space, but that doesn't stop us from using the colorful RGB space to explain the various concepts of vector spaces. When introducing different bases, this chapter also shows you all sorts of parallelogram grids — by now a signature move of this book.

System of Linear Equations

Also unlike many linear algebra textbooks, this book places systems of linear equations last. A system of linear equations is treated as an application case for linear algebra concepts such as matrix multiplication, determinants, and inverse matrices. This chapter again takes a geometric view, first treating a two-variable linear equation as a line and exploring the characteristics of a linear system's solutions. It then further analyzes systems of linear equations using the third perspective on matrix multiplication (linear combinations of column vectors). The chapter closes by introducing elementary row operations for solving systems of linear equations and finding inverse matrices, again blending in various geometric perspectives throughout the explanation.

Now, let's officially begin our journey through Linear Algebra Made Easy and Visual, and make linear algebra truly not hard!